r/DefendingAIArt

For Todd's reeing about "ai is bad for the environment"

Posted by u/madamadam158 • • 0 points • 2 comments

Here is his ai is actually helping the environment

1. Enhancing environmental monitoring
Tracking Deforestation: AI algorithms analyze satellite imagery and data from various sources (including drones) to detect and monitor deforestation in real-time, helping to identify illegal logging activity and assess carbon storage in forests.
Wildlife Monitoring & Anti-Poaching: AI-powered sensors, cameras, and drones can help track wildlife populations, detect anomalies, and identify potential poaching activities, enabling conservationists and law enforcement to respond quickly and protect endangered species.
Pollution Monitoring and Control: AI-driven systems monitor air and water quality, track pollution levels, identify sources of pollution, and provide early warnings for environmental hazards, facilitating timely intervention and improved public health.
Climate Modeling and Prediction: AI algorithms can analyze vast datasets to create more accurate climate models and predictions, helping scientists understand climate change patterns, forecast extreme weather events, and inform adaptation and mitigation strategies.
2. Optimizing energy consumption
Smart Grids: AI can enhance grid efficiency by balancing energy supply and demand, integrating renewable energy sources seamlessly, and minimizing transmission losses.
Energy Efficiency in Buildings & Industries: AI-driven systems analyze energy consumption patterns within buildings and industrial facilities, identify areas of waste, and suggest improvements to reduce overall energy usage and carbon footprint.
3. Advancing sustainable agriculture
Precision Agriculture: AI analyzes data on soil health, crop growth, weather patterns, and more, allowing for precise application of water, fertilizers, and pesticides. This optimizes resource use, reduces chemical runoff, and improves crop yields.
Early Disease and Pest Detection: AI-powered tools like image recognition and drones can detect crop diseases and pests in their early stages, enabling timely interventions and minimizing pesticide use.
4. Facilitating efficient waste management
Improved Recycling: AI can help analyze waste processing and recycling facilities to recover and recycle more materials and optimize waste collection and logistics, reducing emissions associated with waste management.
Reducing Food Waste: AI can be used to predict inventory needs and optimize supply

chains in the food industry, reducing food waste at various stages, from farming to consumer consumption.
5. Supporting sustainable urban development
Smart City Planning: AI can help optimize resource utilization, streamline traffic flow, and improve urban planning decisions, leading to more sustainable and liveable cities.
In essence, AI offers the ability to analyze vast amounts of data, predict outcomes, and automate processes, enabling more efficient and effective environmental management and sustainability efforts. This includes enhancing environmental monitoring, optimizing resource consumption in energy and agriculture, improving waste management and supporting sustainable urban development. AI-driven solutions have reduced energy consumption by 10-60% in certain sectors, according to the World Economic Forum.
Optimizing factory processes using AI can lead to energy consumption reductions of 30-50% according to a study published in Environmental Chemistry Letters cited by Yale University.
Widespread adoption of AI in the manufacturing sector alone could lead to energy savings equivalent to Mexico's total energy consumption today, notes Orennia.
AI-controlled HVAC systems in buildings could reduce global electricity demand by 300 TWh annually, according to the IEA.
Specific examples include:
A pharmaceutical company saved 156,000 kilowatt-hours of electricity annually by using AI to optimize equipment efficiency.
The Wellcome Sanger Institute significantly reduced energy consumption for genomic analysis, saving approximately 1,000 megawatt-hours annually and potentially reducing costs by $1 million compared to traditional CPU-based methods.

Comments

u/Drakahn_Stark • 3 points • 2025-08-12 18:59 UTC

I would add that a lot of AI companies are already fixing older sustainability problems.

Google and Microsoft, for example, are building carbon neutral and zero waste data centres powered by clean energy like renewables and nuclear, solving high energy waste in computing from before AI.

> **u/[deleted]** • 1 points • 2025-08-12 22:17 UTC
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u/Froggyshop • 3 points • 2025-08-12 15:41 UTC

If only antis could read they would be very angry right now.